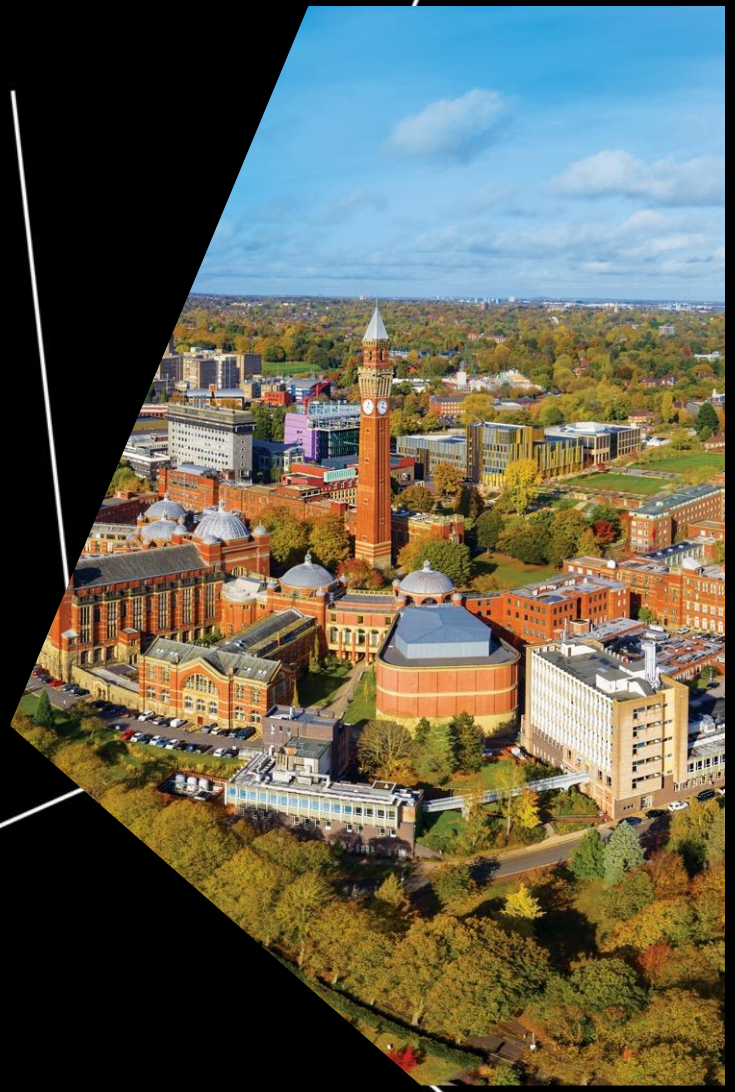
Module: FON

Reflex

Lecture : 9

Dr Ratnadeep Saha
Email: r.saha@bham.ac.uk



UNIVERSITY OF BIRMINGHAM

Learning outcome:

Understanding of multi-synaptic pathways of reflexes and their importances

Need to know:

- What is reflex and their components
- Receptors involved
- Type of reflexes
    - Stretch reflex (Basic mechanism)
    - Golgi Tendon reflex
    - Withdrawal reflex
    - Cross extensor reflex

Recap – peripheral nervous system

CNS = Brain + Spinal cord

PNS = Nerves + Ganglia (+ receptors/plexuses)

Peripheral nervous system

(**Movement control** + Process sensory input)

Somatic nervous system

Motor (efferent) system

Somatic motor

Skeletal muscles contraction

Sensory (afferent) system

Somatic sensory

Sensory receptors e.g. touch, hot, pain etc

Autonomic nervous system

Sympathetic division

Parasympathetic division

Visceral motor

Smooth muscles, cardiac muscles, glands

Visceral sensory

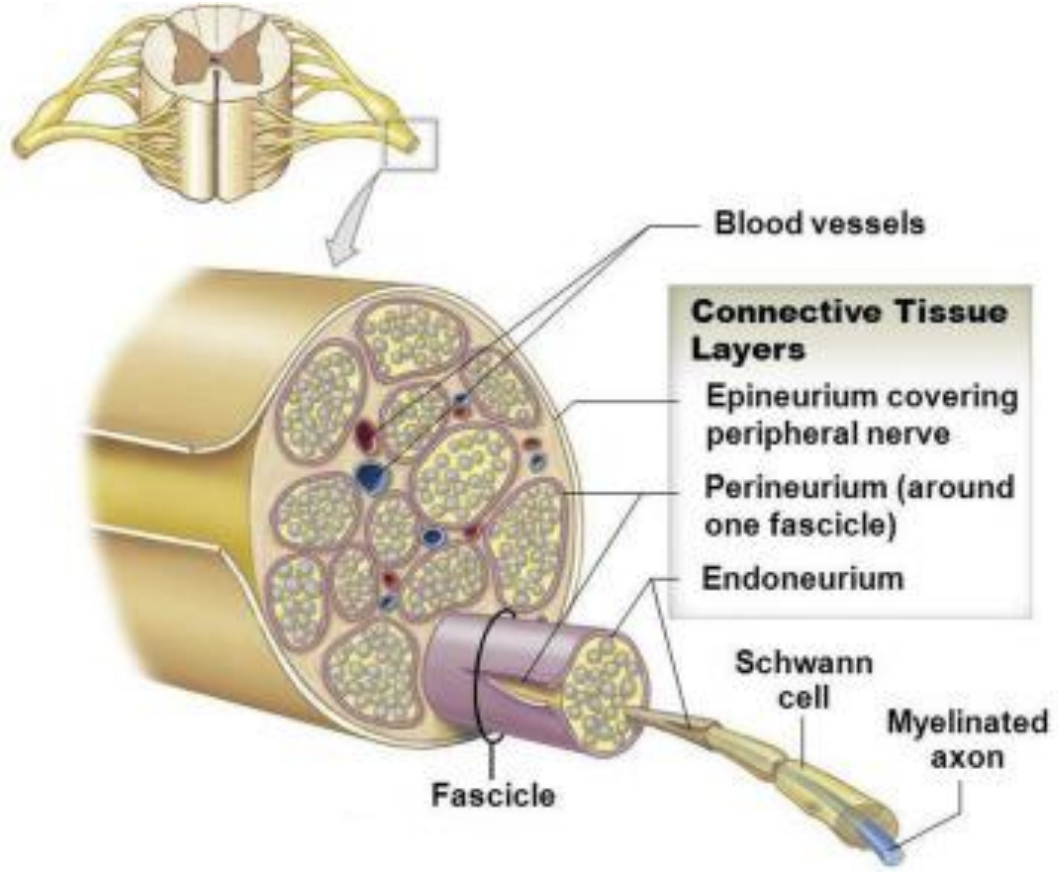
Visceral organ e.g. abdominal pain, Pupil size changes, BP changes

- **Movement control**: In initiating & controlling the movements of your body in liaison with motor system.
- **Sensory input**: Processing sensory information via external stimuli. (Proprioception, touch, pressure, temperature, pain).

Reiteration :

Sensory (afferent) VS Motor (efferent)

- Peripheral nerves include the spinal nerves and cranial nerves
- Cranial nerves: 12 pairs (some purely sensory / motor /mixed type)
- Spinal nerves: 31 pairs (Only Mixed type)
- Nerves are bundle of neurons
- Covers in layers of connective tissue

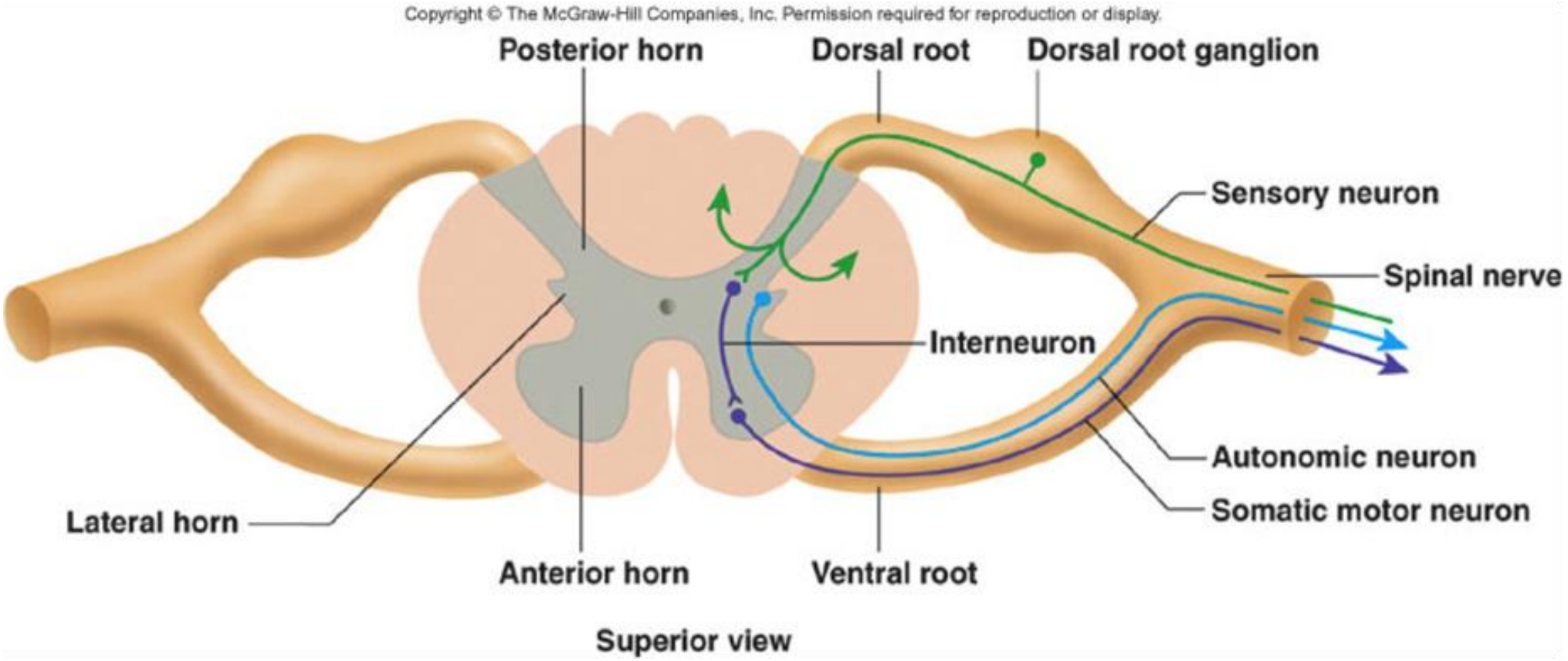


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Reiteration :

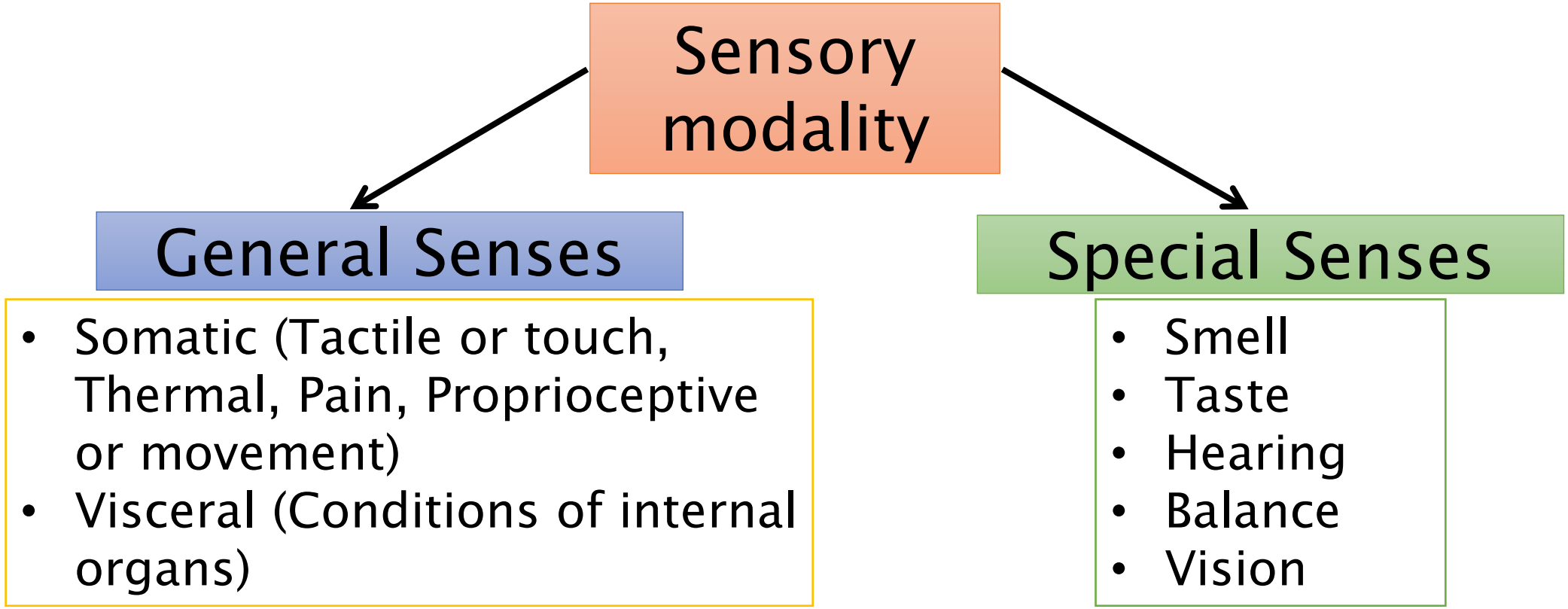
Sensory **A**fferent **M**otor **E**fferent

Afferent = **A**rrive / **E**fferent = **E**xit



Sensory (afferent) VS Motor (efferent)

- To make it more complicated, the nerve fibres can be further categorized into *general* or *special*. So, you have:



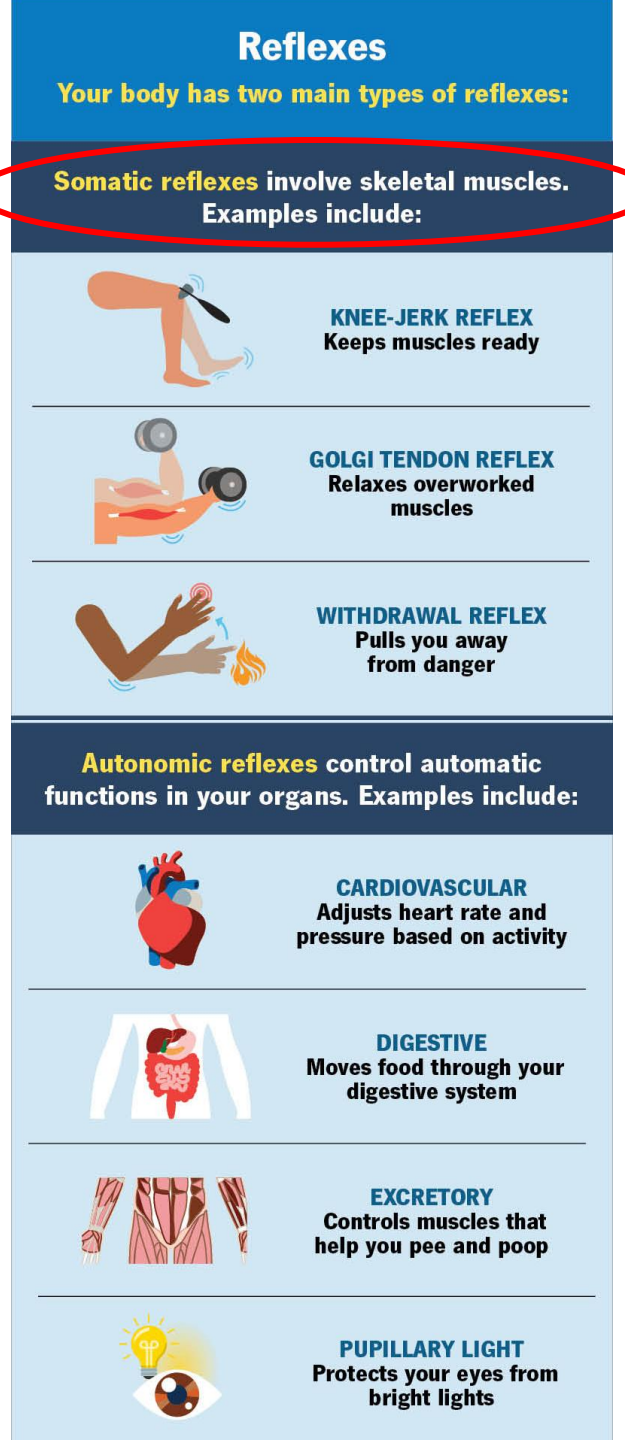
Note: Special senses are processed via cranial nerves and differ from the pathway utilized in processing general somatic senses. (General somatic : Discussed in Somatosensory session)

Reflex (A type of Motor activity)

is automatic, involuntary response

& is also a class of movement.

Let's see briefly what are the **types of movements** are and what **motor neuron type** is principally involved in it.



Types of Movement:

1.Reflex

- ✓ **Withdrawal reflex**
- ✓ **Stretch reflex**
- ✓ **GTO reflex**

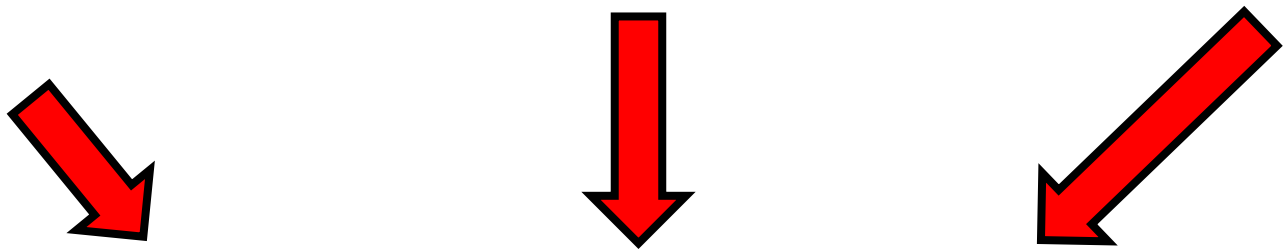
2. Rhythmic

- ✓ **Swinging of hand**

3. Voluntary

Any kind of action

Decided by brain



Alpha motor neurons

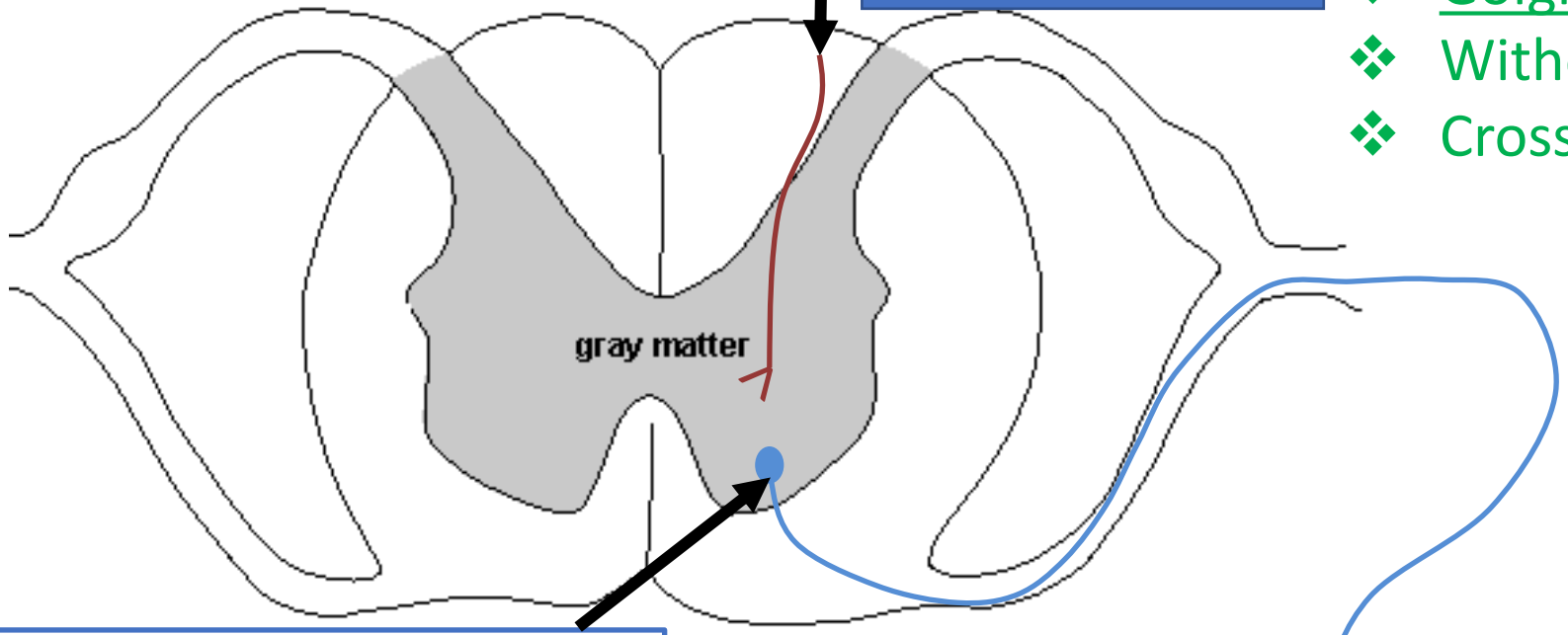
Alpha-motor Neuron

Arises from anterior horn cells of spinal cord, passes via ventral root & connects extrafusal fibers (Skeletal muscle).

Connected with **gamma motor neuron** (Not shown here) which innervates intrafusal muscle fiber of skeletal muscle (See later)

Upper Motor Neuron (Comes from brain)

DORSAL/POSTERIOR



Lower (alpha) Motor Neuron

VENTRAL/ANTERIOR

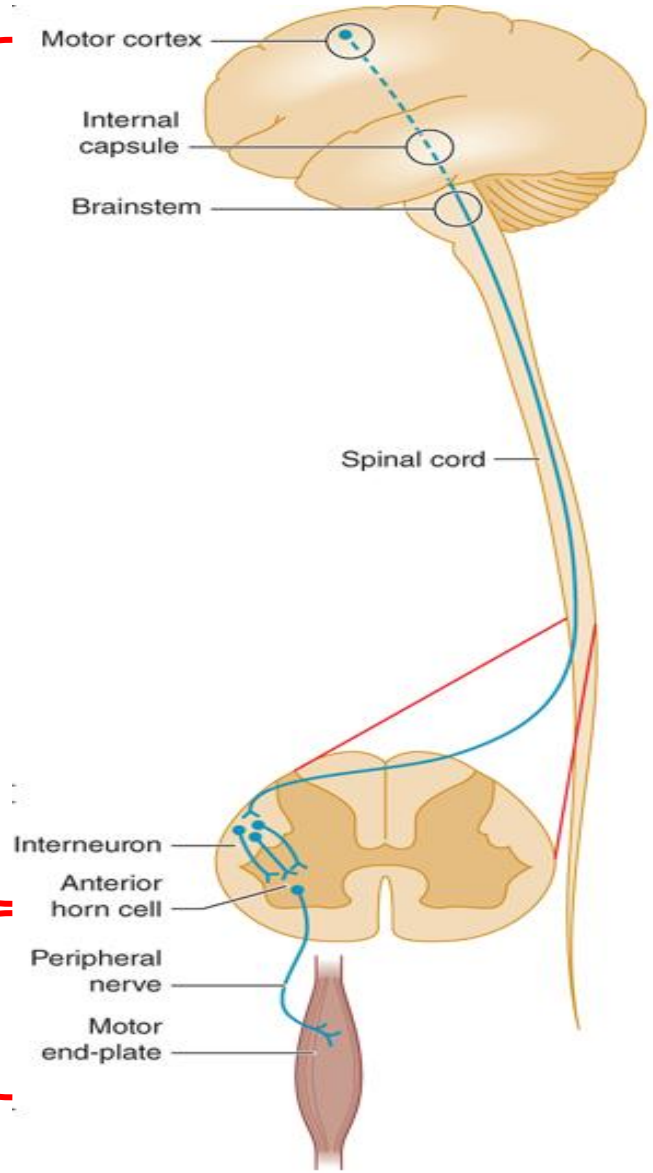
Muscle

Linked to reflexes like:

- Stretch reflex
- Golgi Tendon reflex
- Withdrawal reflex
- Cross extensor reflex

α- Motor neurons are directly responsible for initiating muscle contraction & sensing changing muscle length

BIGGER PICTURE



Upper motor neurones

Lower motor neurones

J. Aminoff, D. A. Greenberg, R. P. Simon: Clinical Neurology, 9th Edition
smedicine.com
© McGraw-Hill Education. All rights reserved.

Diagram showing how motor neurons at different levels are connected

Motor neurons control the contraction of skeletal muscles and are the final common pathway. (controls motor behaviour)

Muscles become paralyzed when their innervating motor neurons die because of injury or disease. Eg: Amyotrophic Lateral Sclerosis, progressively destroy motor neurons (both upper and lower)

What is a reflex?

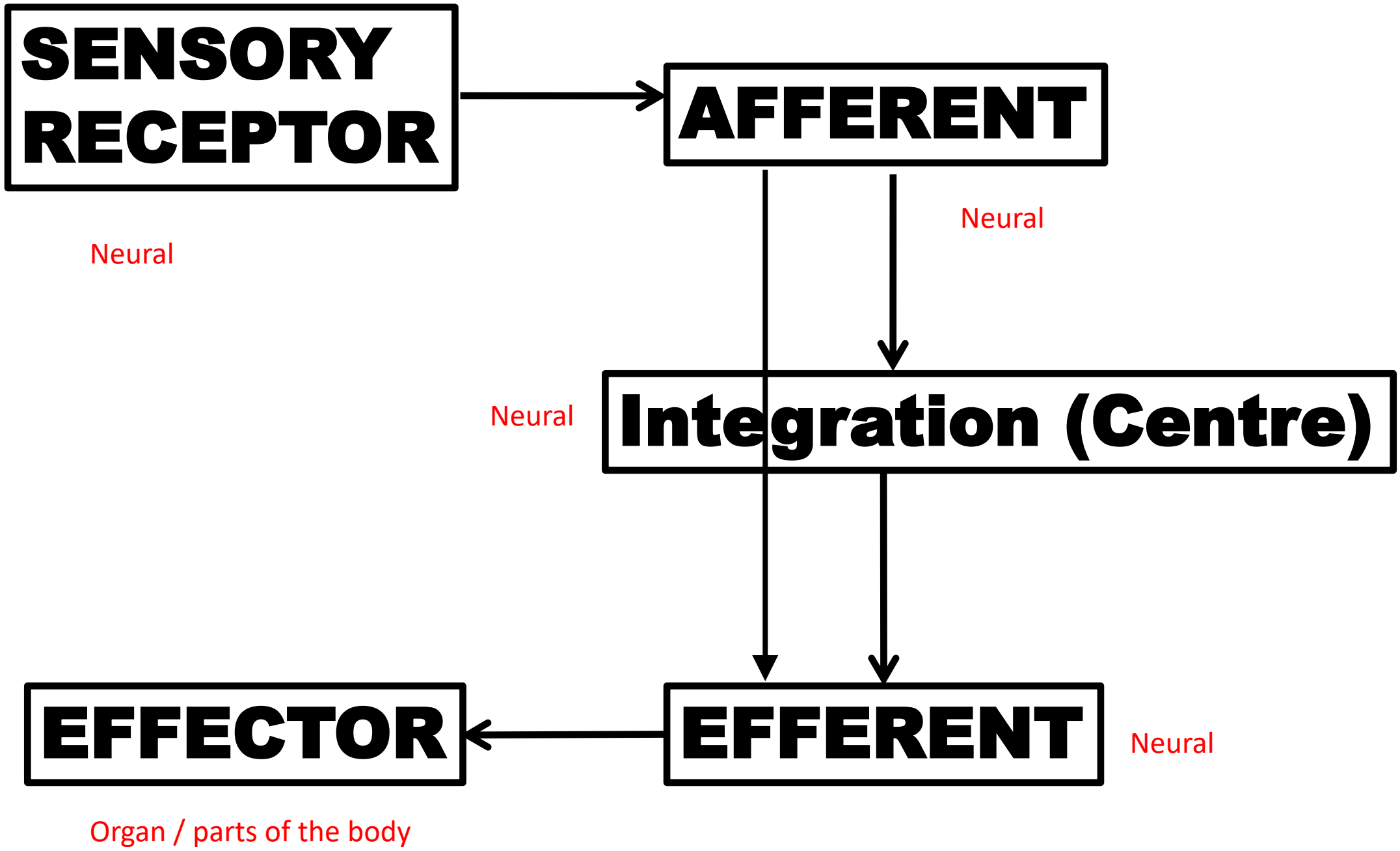
- Stereotyped
- Involuntary
- Rapid

General functions of reflexes

1) protective

2) postural control

3) Homeostasis


SENSORY RECEPTOR
AFFERENT
Neural
Neural
Neural
Integration (Centre)
EFFECTOR
EFFERENT
Neural
Organ / parts of the body

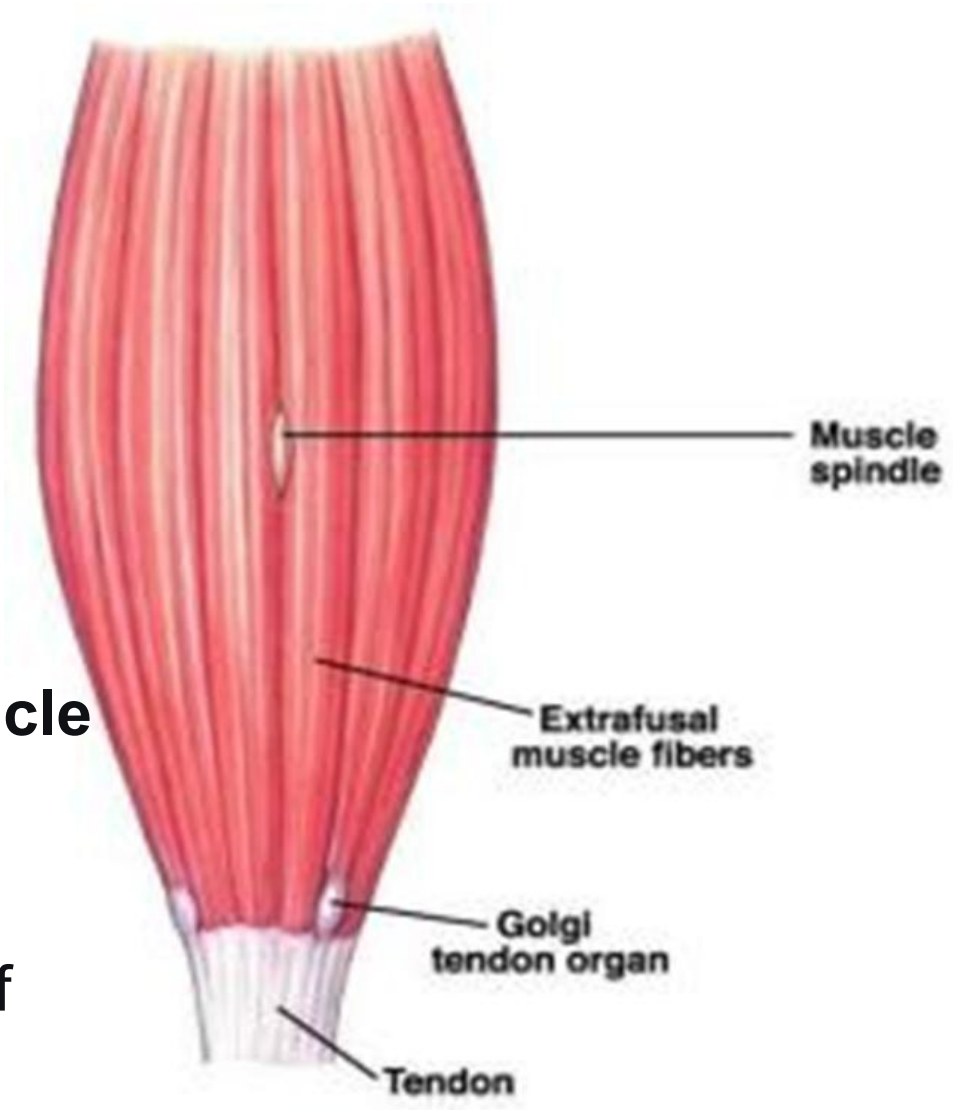
Stretch reflexes (Myotatic)

- Adjust degree of muscle contraction (See next)
- Posture

- Involve sensory receptors (mainly Proprioceptors) & the Effector organs

Proprioceptors (Location)

- Muscle spindle
- Golgi tendon organ



Note:

Extrafusal muscle fibres = standard **skeletal muscle fibres** .

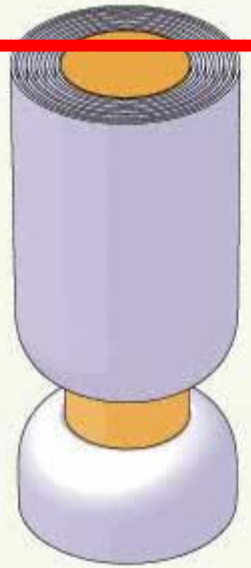
•They are innervated by **alpha motor neurons (Efferent)**, and extrafusal fibres make up the bulk of skeletal muscle

Proprioception

- Detects the mechanical status of the musculo-skeletal system

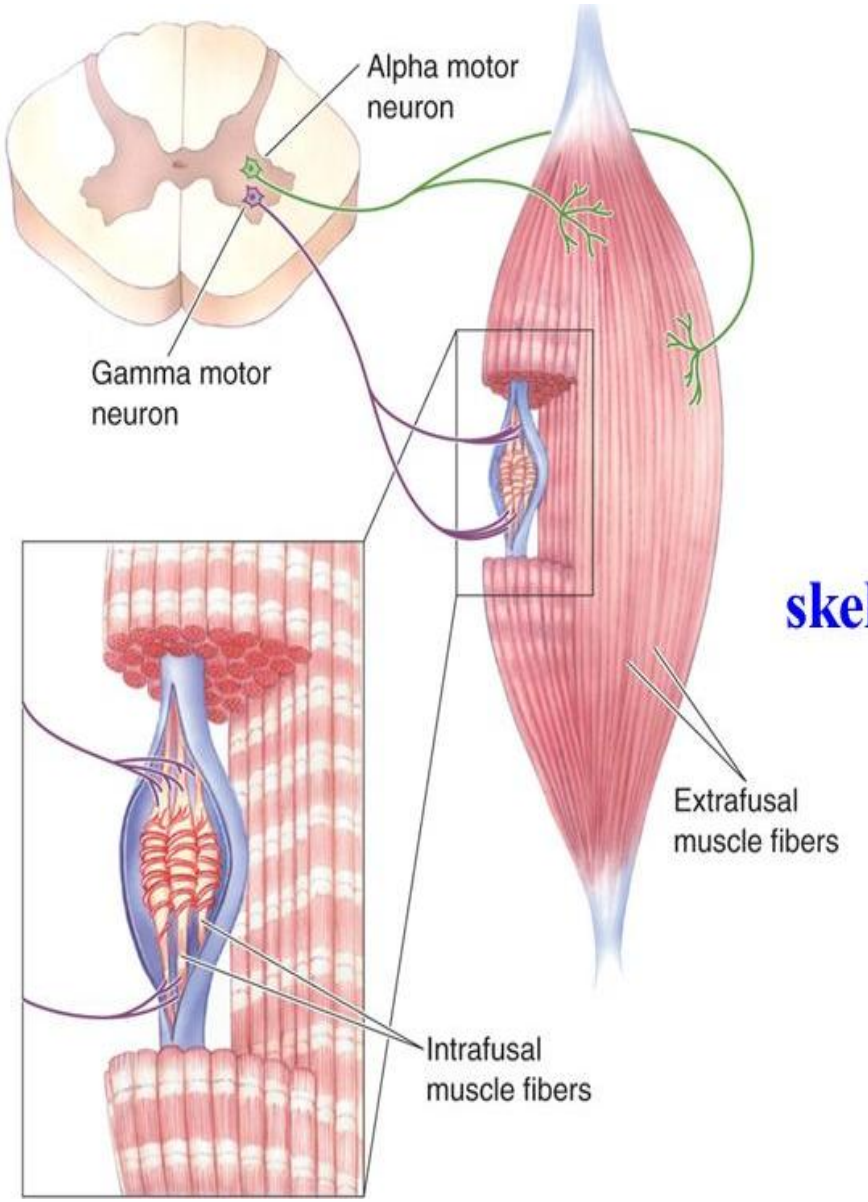
- Provide information about:
  - joint position
  - muscle length
  - muscle movement
  - acceleration
  - tension / force
- Proprioceptors involved in **Simple reflexes by Aα type (Gr I) afferent fibres**

Primary afferents – different axon classifications

Axons from skin	Aα	Aβ	Aδ	C
Axons from muscles	Group I	II	III	IV
				
Diameter (μm)	13–20	6–12	1–5	0.2–1.5
Speed (m/sec)	80–120	35–75	5–30	0.5–2
Sensory receptors	Proprioceptors of skeletal muscle	Mechanoreceptors of skin	Pain, temperature	Temperature, pain, itch

Other types we will discuss again in Somatosensory SGT session

Neuroscience, Exploring the Brain. 3rd edition Bear Connors & Paradiso



The extrafusal muscle fibers are innervated by Alpha motor neuron

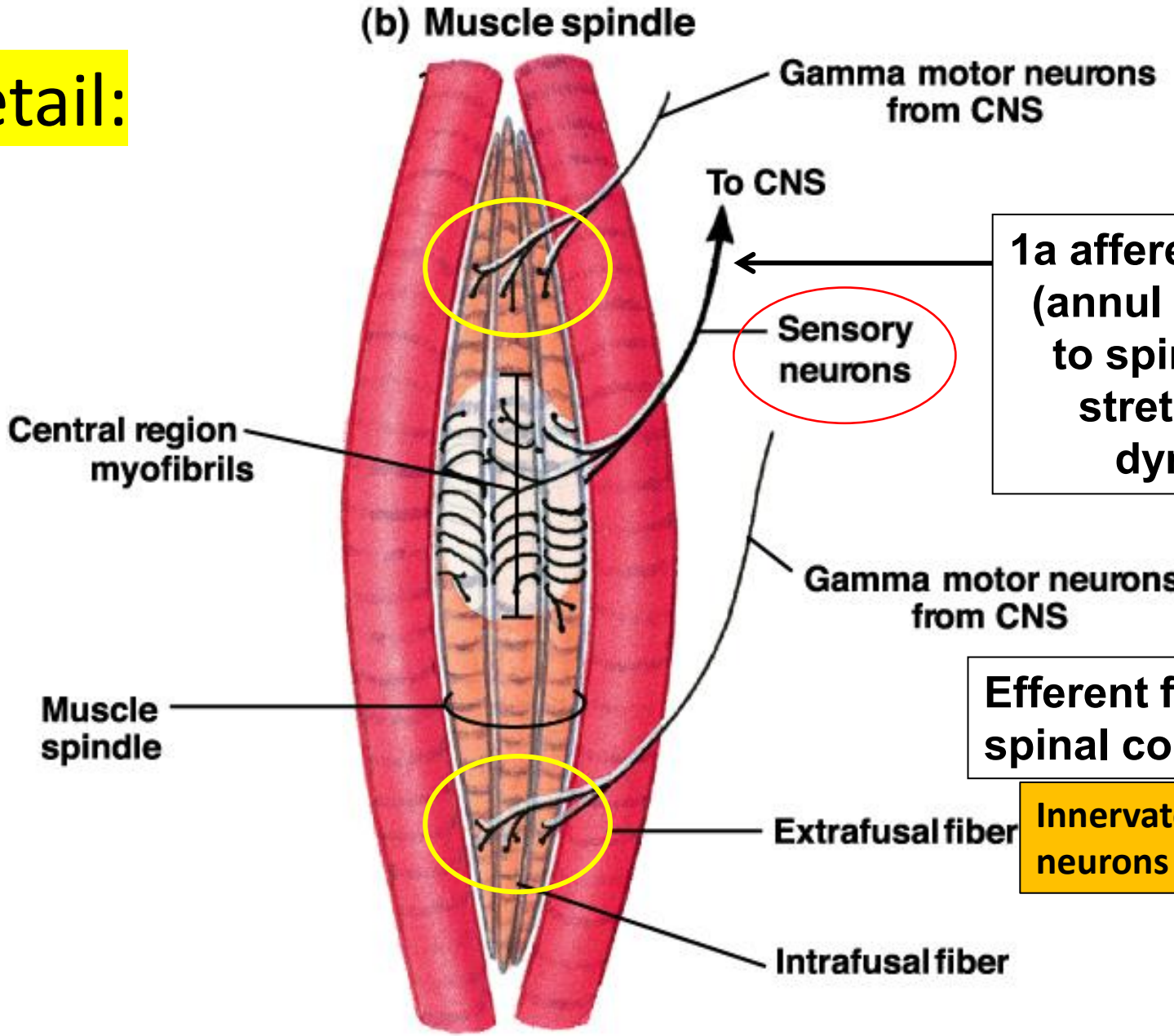
skeletal muscle contraction

The intrafusal muscle fibers are innervated by Gamma motor neurons

Fig: Extra and intrafusal muscle fibre and **Muscle spindle**

(Eg: Spindle through intrafusal fibres regulate sensitivity [eg: amount & rate] of muscle to stretch)

Greater detail:



1a afferent: Primary sensory (annul spiral ending) goes to spinal cord (Tied with stretch reflex): Detect dynamic activities

Efferent from spinal cord

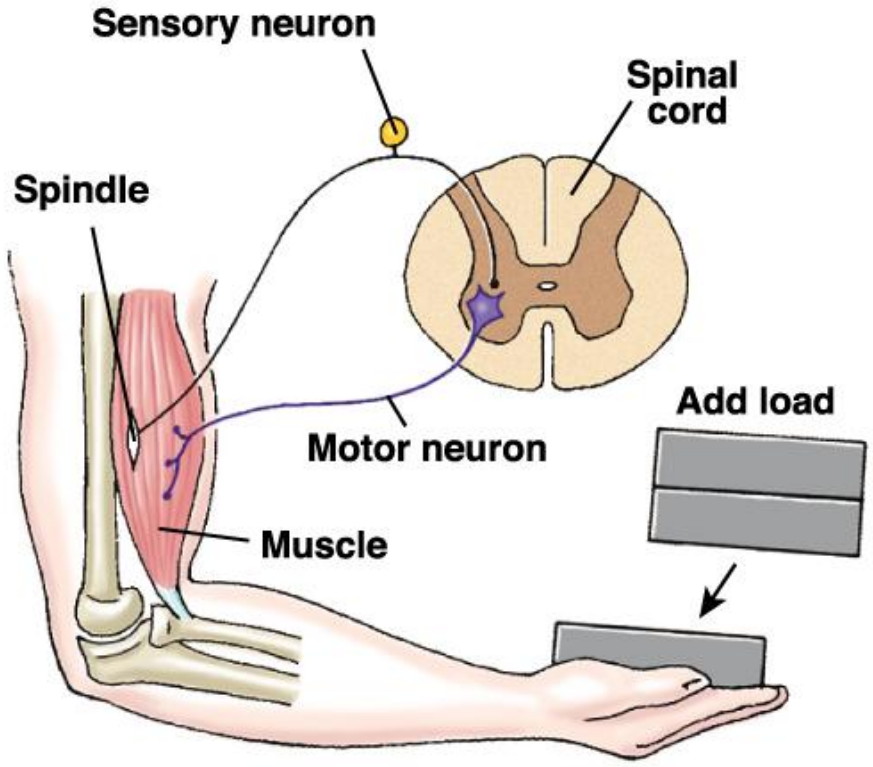
Innervated by alpha motor neurons (Not shown here)

1. Adjust degree of muscle contraction / load : How?

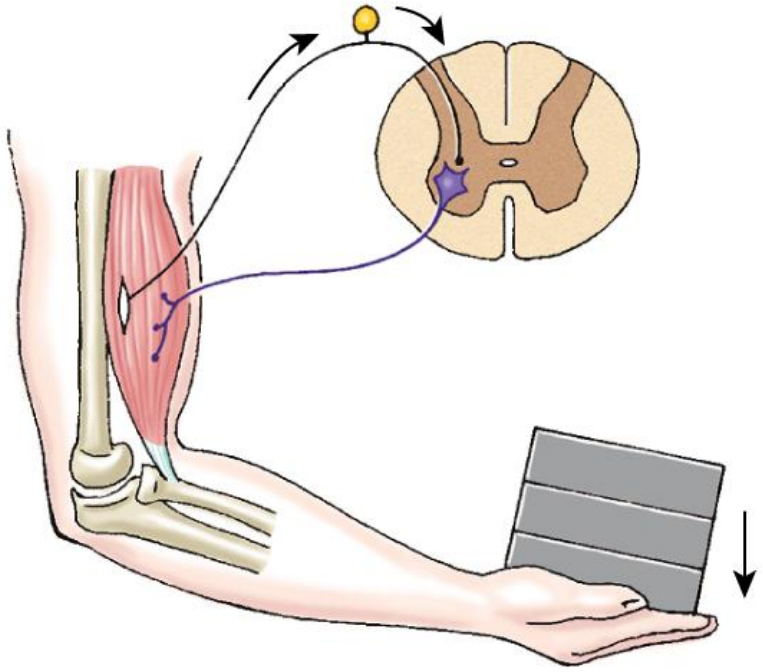
2. What is the role of Gamma motor neuron?

Adjust degree of muscle contraction / load: How?

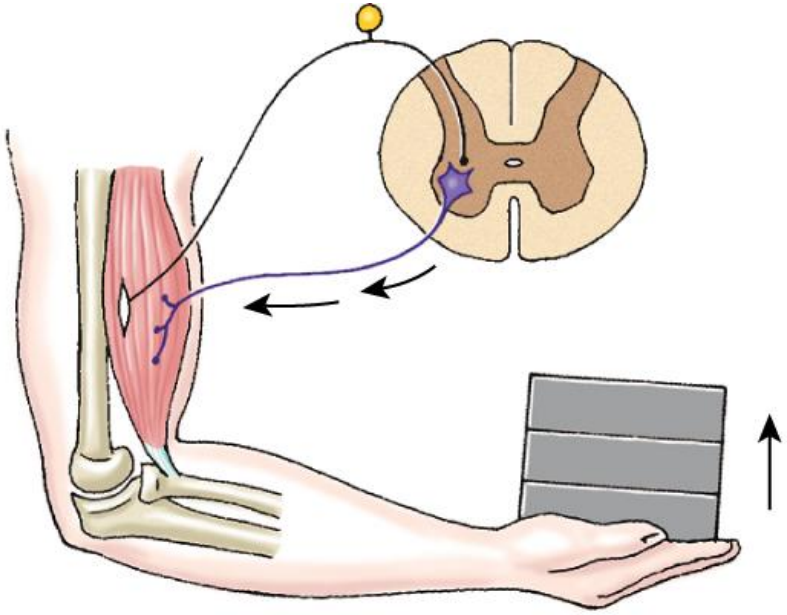
Stretch reflex ; monosynaptic to agonistic muscle (Biceps brachii) - maintains contraction with increased load within a reasonable limit



(a) Add load to muscle

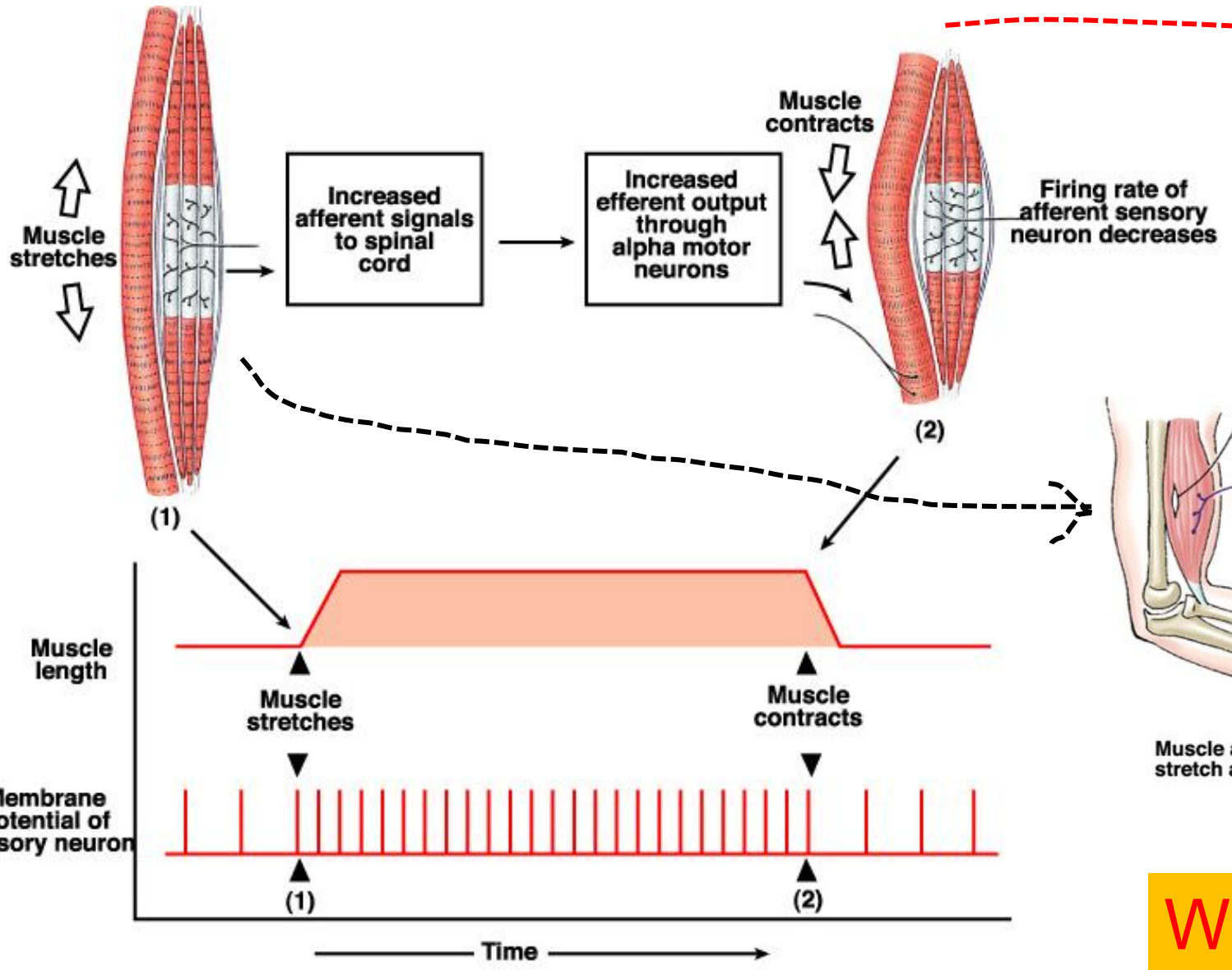


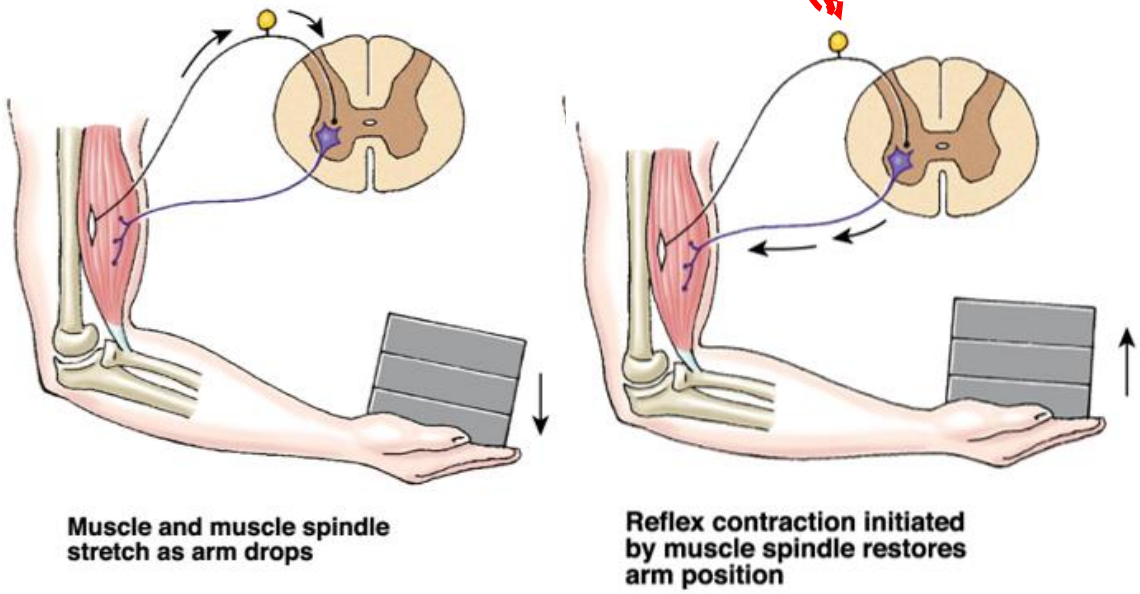
Muscle and muscle spindle stretch as arm drops



Reflex contraction initiated by muscle spindle restores arm position

Response of muscle spindle to stretch (In general)





Muscle and muscle spindle stretch as arm drops

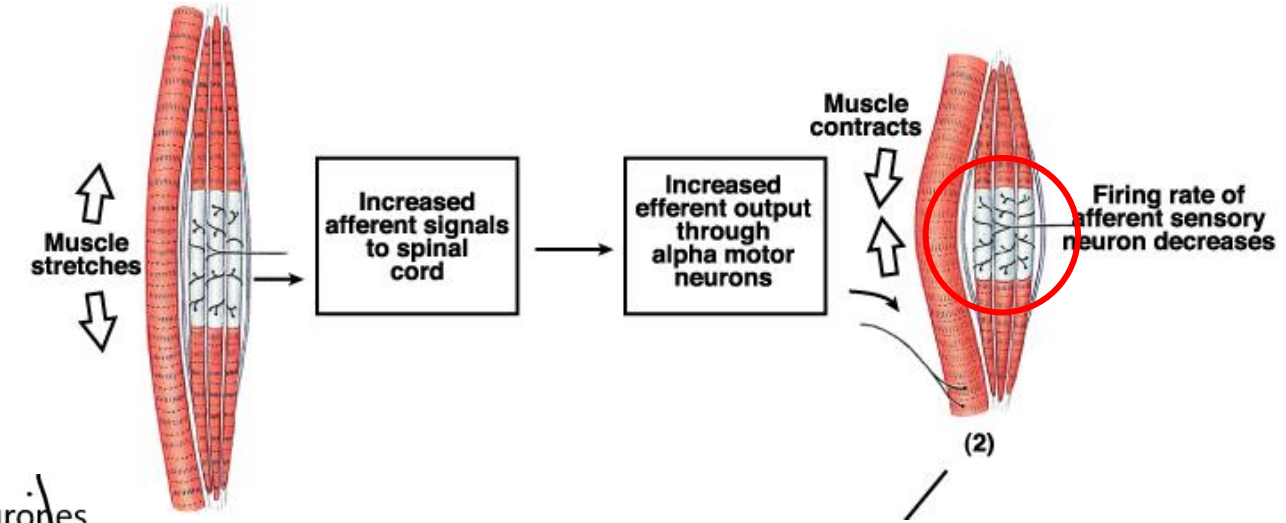
Reflex contraction initiated by muscle spindle restores arm position

What is the role of Gamma motor neuron? [See next]

Explanation on previous slides (How: Adjusting muscle contraction) :

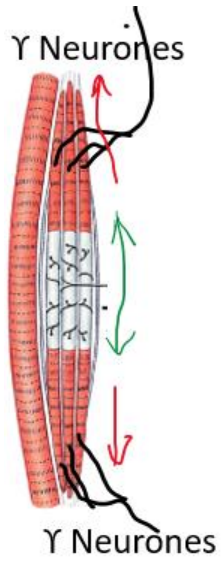
Activated spindles → sensory information to spinal cord via sensory neurons (Ia type) → directs back via alpha motor neuron to extrafusal muscle fibers (not shown here) → relaxes the center of the spindle → stop firing (by sensory neuron 1a). This is functionally *disadvantageous* and hence we need extra gamma connectivity.

Why unloading disadvantageous: Stopping firing is called **unloading** → functionally disadvantageous as brain stop receiving signals from periphery (rate of shortening, length etc.)



How to overcome:

Counter mediated as we have additional gamma motor neurons that innervate the tapered end → contraction → lengthens the central region and maintain stretch instead.



Gamma Importance: Adjust the stretch sensitivity of the muscle spindles so that spindles are active no matter what the muscle length is.

Note: Gamma is *technically working always in the background*.

Stretch reflex usage (For Posture explanation)

Stretch reflex (myotatic reflex) : major stimulus is **stretching**.

A major role of the myotatic reflex is the maintenance of ***posture***. If one is standing upright and starts to sway to the left what would happen?

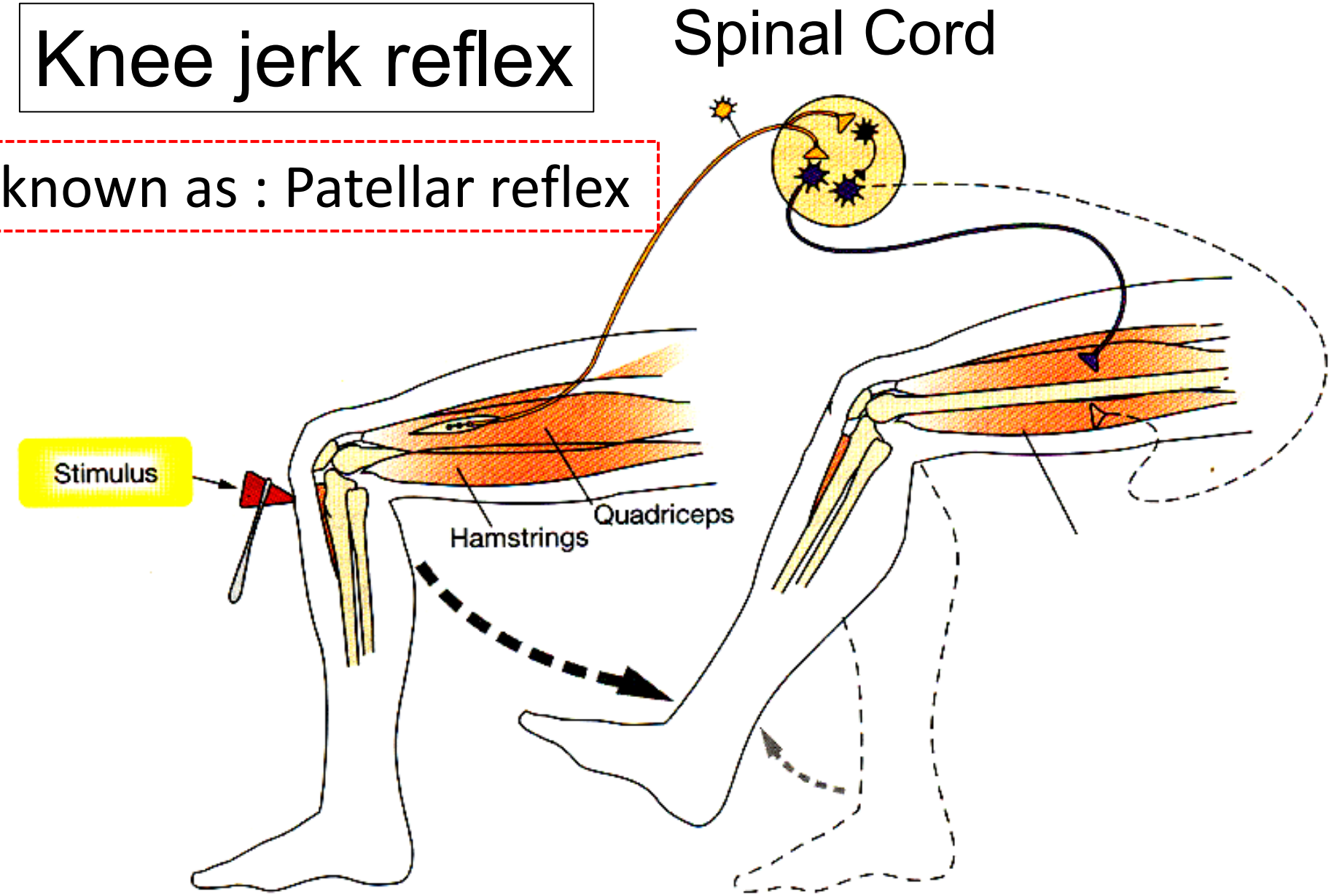
The muscles that **counteract** the sway are on the **opposite side** to the direction of sway.

Sway **left** → **right** muscles (Right hip abductors stretched → Begin **right** stretch reflex → pull back (Contraction) **right** → posture restored.

One example of myotatic reflex is Knee jerk reflex, follow the next slides write up:

Knee jerk reflex

Also known as : Patellar reflex

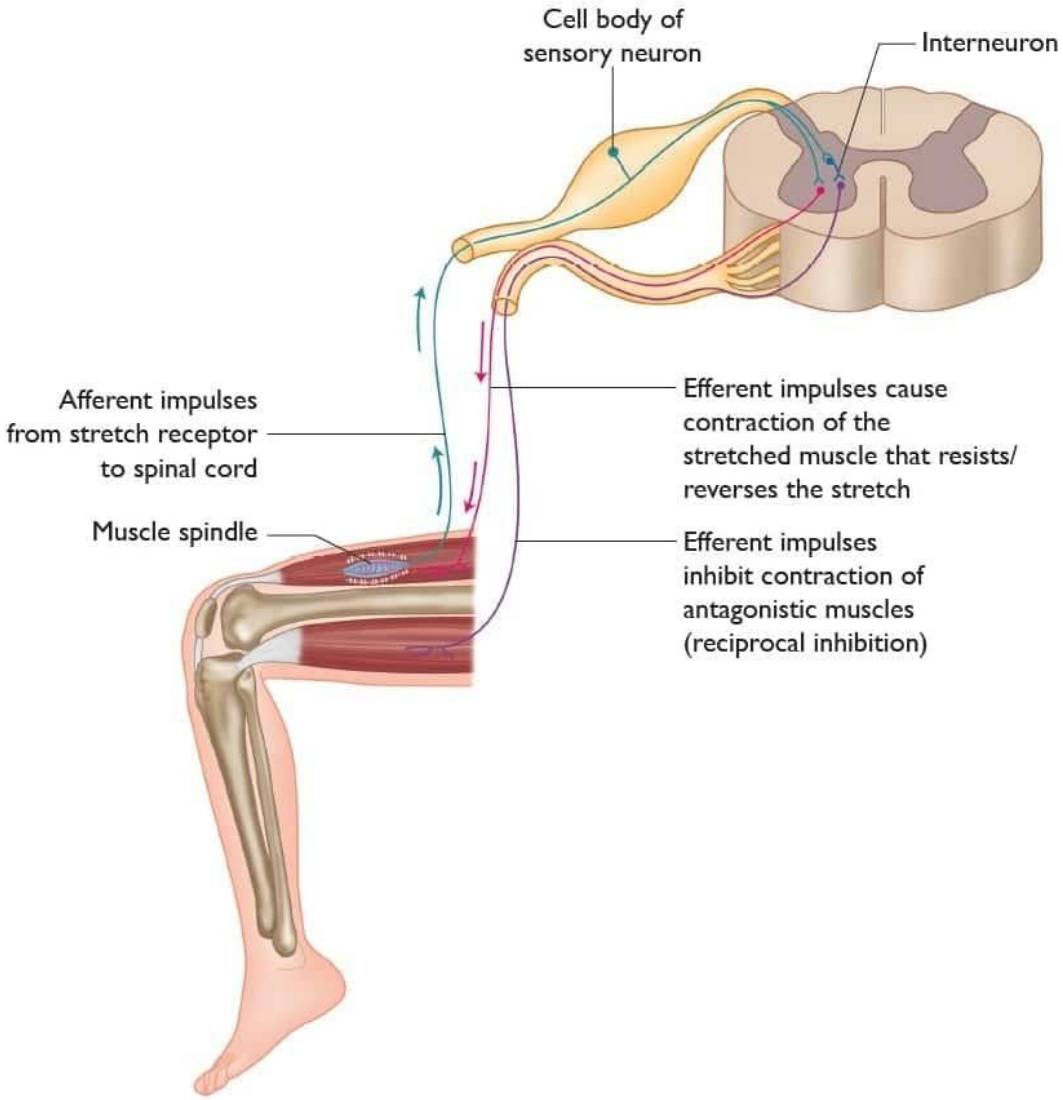


From: Silverthorn: Human Physiology. Pearson

Stretch reflex: α motor neuron

(Myotatic reflex; E.g. Knee jerk reflex)

Stimulus: **Stretch of muscle** (Length)



1) When taps the patellar tendon with a hammer, this action causes the knee extensor muscle to stretch abruptly.
2) This stretch activates the myotatic reflex, reflecting an extension of the lower leg (deep tendon reflex)

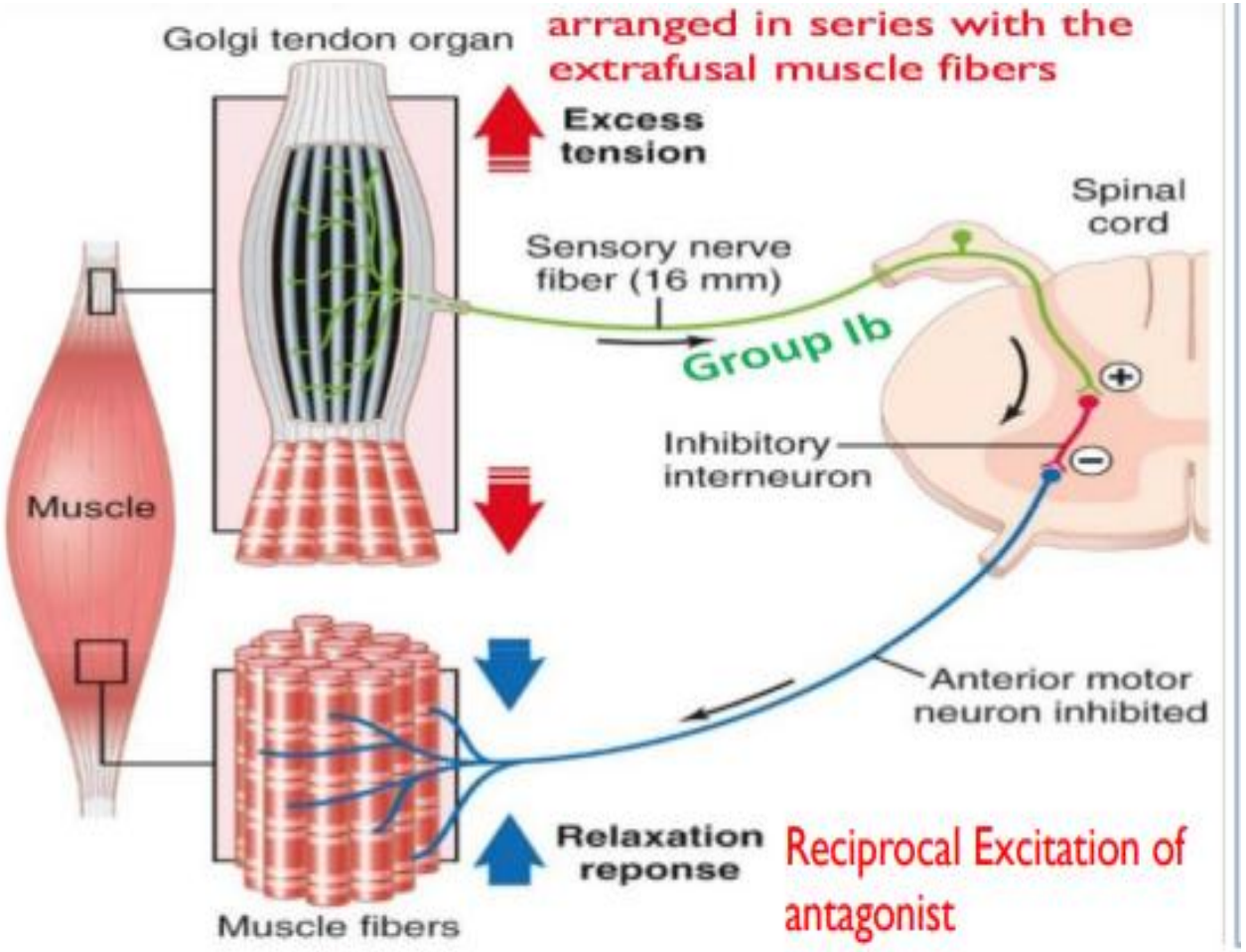
3) This is initiated by the Muscle spindle, not the Golgi Tendon Organ.

4) Sensory information ----Ia afferent fibres that connect α-motor neuron----- to produce contraction in the same muscle.

5) A branch of same sensory neuron ----via an inhibitory neuron connect to an α motor neuron that innervates (opposite) antagonistic muscle (Hamstring muscle) to induce inhibition (relax). (Reciprocal inhibition)

Golgi tendon reflex: α motor neuron
(Inverse stretch reflex)

Stimulus: Tension / change of tension in muscle due to **overload**

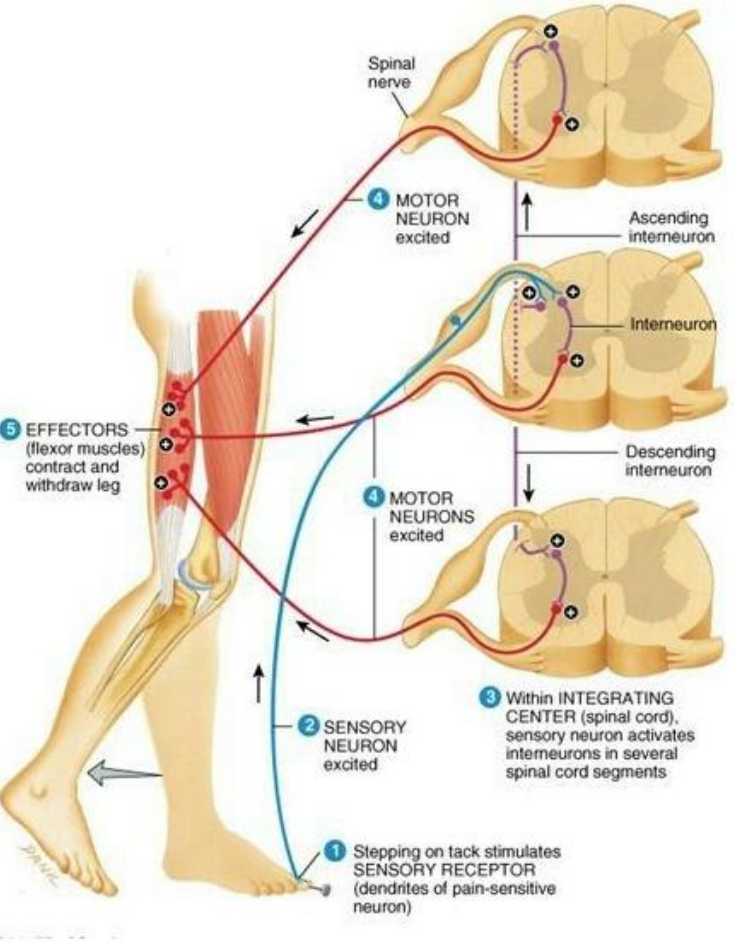


1) Tension / Change of tension at muscle
2) GTO activated
3) Send sensory signal via Ib afferent fibers to inhibitory interneuron
4) Inactivates α motor neuron that inhibits extrafusal muscles of the same muscle gr
5) Relaxes (Lengthening reaction)

Important: <u>Prevent tearing of muscle</u>

Gr Ib sends excitatory signals to a α motor neuron (innervating antagonist muscle)…induce contraction (Reciprocal excitation)

Flexor (withdrawal) Reflex



- Step on tack (pain fibers send signal to spinal cord
- Interneurons branch to different spinal cord segments
- Motor fibers in several segments are activated
- More than one muscle group activated to lift foot off of tack

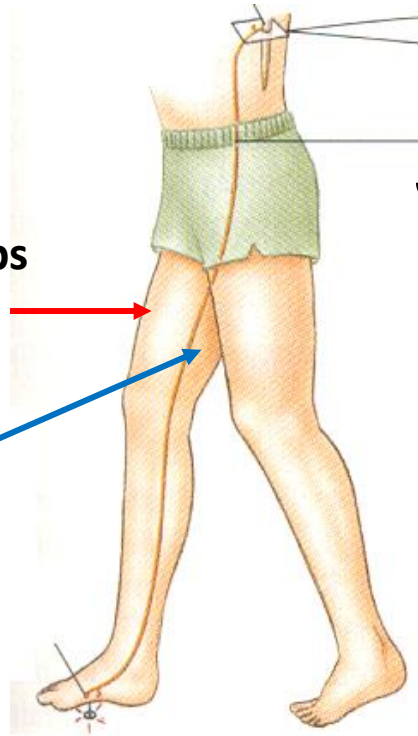
1) Can be initiated by nonproprioceptive receptors as well as by proprioceptors.

2) Sensory fibers are gr III afferent

3) Protective reflex

Often experience this reflex after accidentally touching a hot stove or a sharp object, as we withdraw our hand even before we consciously experience the sensation of pain.

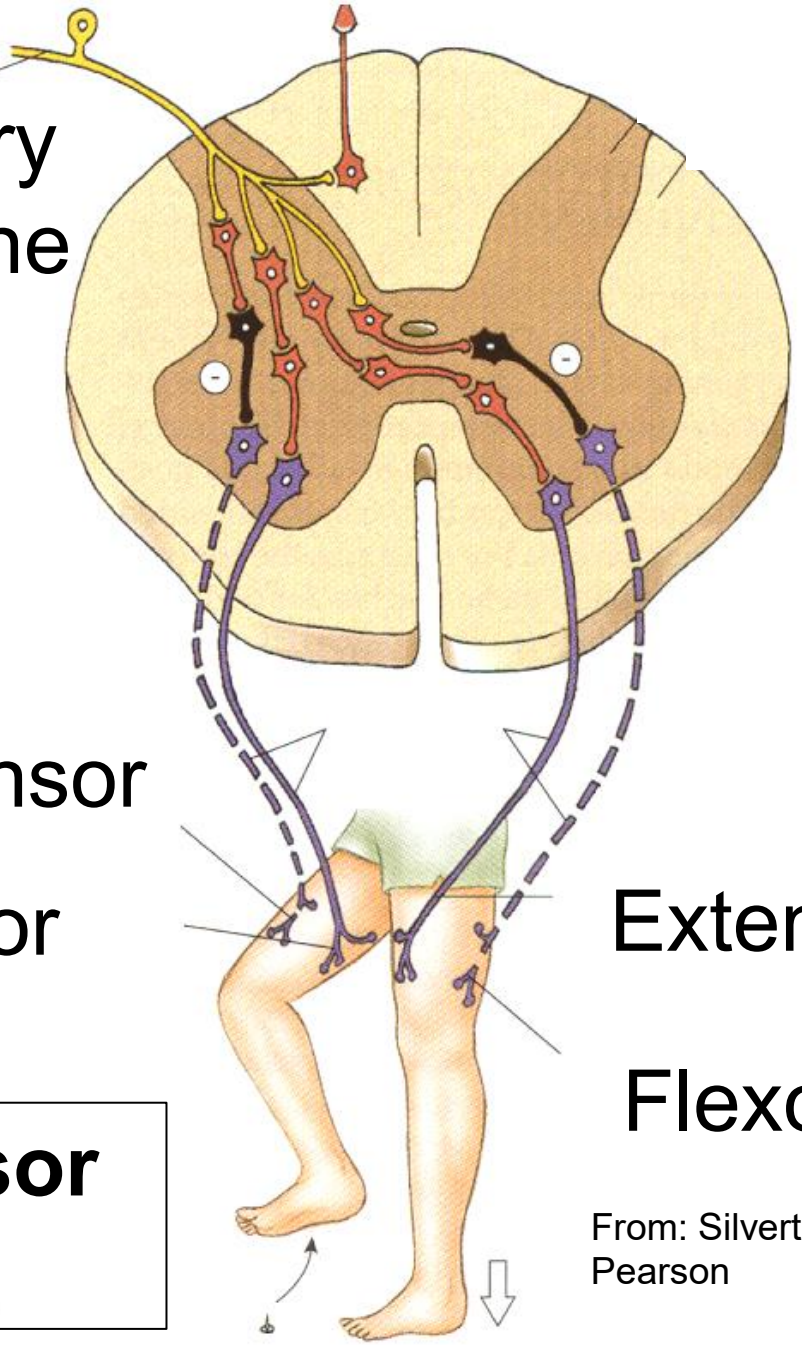
4) Activates large number of muscle groups of different segments



Quadriceps
(Flexor)

Hamstring
(Extensor muscle)

Sensory Neurone



Extensor

Flexor

Extensor

Flexor

Crossed Extensor Reflex

From: Silverthorn: Human Physiology. Pearson

Crossed Extensor Reflex continues…..

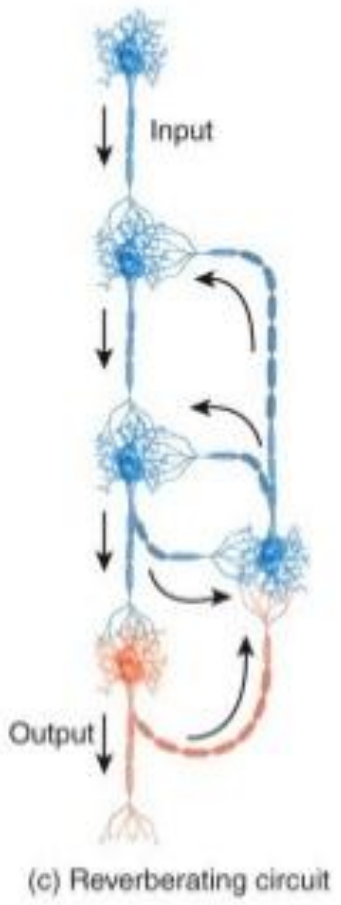
- Occur simultaneously, coordinated with flexor reflex

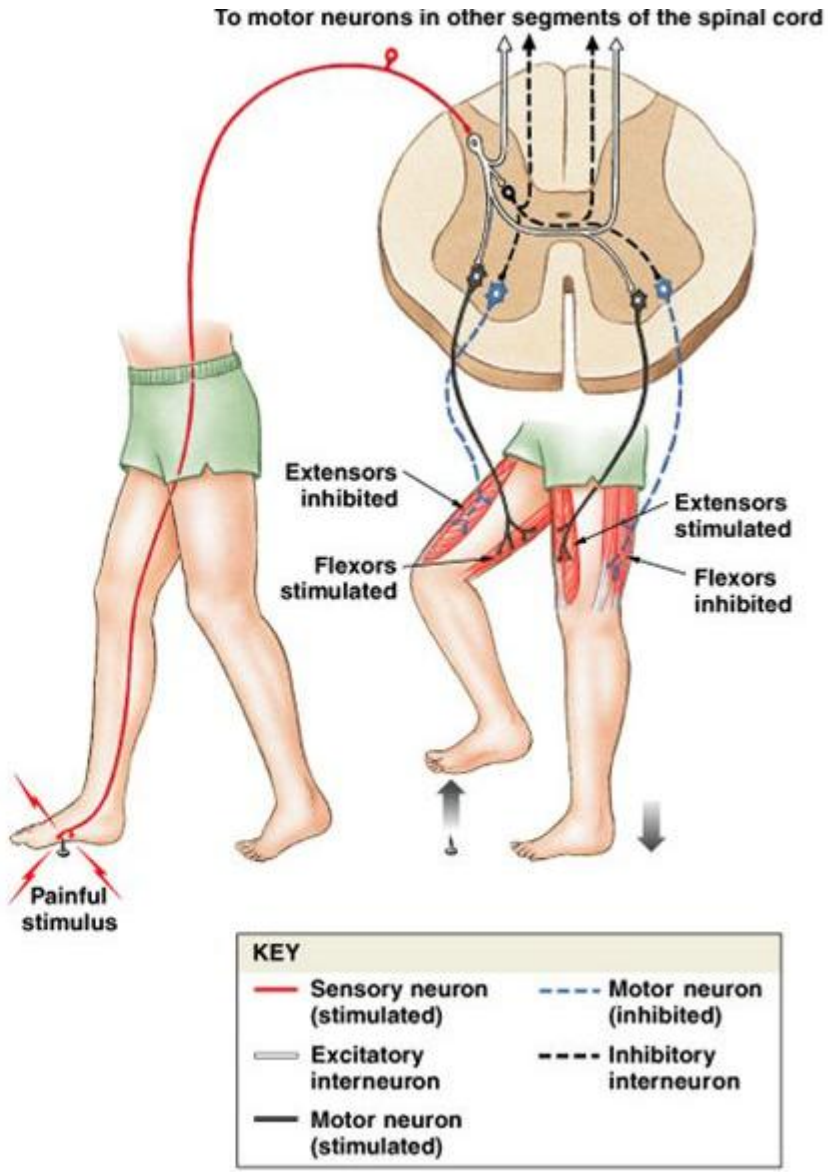
  e.g., flexor reflex causes leg to pull up:

  – crossed extensor reflex straightens other leg

  – to receive body weight

  – maintained by reverberating circuits





Major Impacts (lower motor neuron damage):

Muscle atrophy: Deprived of necessary trophic factors

Weakness: Damage to alpha motor neurons and the atrophy of muscles

Fasciculation: Damaged α motor neurons can produce spontaneous action potentials. These spikes cause the muscle fibres to fire (Visible response).

Fibrillation: With further degeneration of α motor neuron, remnants remain. These individual axon fibers can generate spontaneous action potentials which only cause individual muscle fibers to contract. Fibrillations are too small to be seen (possible by EMG).

Hypotonia: The loss of alpha motor neurons causes a decrease in muscle tone.

Hyporeflexia: The **myotatic (stretch) reflex** is weak or absent because the alpha motor neurons are damaged.

Important Terminologies :

1. **Reflex:** A rapid, involuntary, stereotyped (predictable) motor response to a sensory stimulus. It is the simplest form of movement and does not require conscious brain involvement, though the brain can modulate it. **Functions:** Protection (e.g., withdrawing from pain), postural control (e.g., standing upright), and homeostasis.

2. **Alpha (α) Motor Neuron:** The large motor neuron located in the anterior (ventral) horn of the spinal cord. Its axon travels via the ventral root to innervate **extrafusal muscle fibres** (the main force-generating fibres of skeletal muscle). **Importance:** It is the **"final common pathway"** for all motor behaviour. All voluntary movements, rhythmic movements, and reflexes ultimately converge on alpha motor neurons to cause muscle contraction.

3. **Gamma (γ) Motor Neuron:** A smaller motor neuron that innervates the **intrafusal muscle fibres** located inside the muscle spindle. **Importance:** It adjusts the **sensitivity** of the muscle spindle to stretch. Without gamma neurons, when a muscle shortens, the spindle would go "slack" and stop firing (a problem called **unloading**). Gamma neurons contract the spindle ends to maintain tension on the central region, ensuring the spindle remains active at all muscle lengths and allowing brain to continuously monitor muscle length.

4. **Muscle Spindle:** A specialized sensory receptor (proprioceptor) located within the belly of a muscle running **in parallel** with the main (extrafusal) muscle fibres. It contains intrafusal fibres and is innervated by both sensory (Ia afferent) and gamma motor neurons. **Function:** Detects **muscle length** and the **rate of change in length** (stretch).

Important Terminologies :

5. Golgi Tendon Organ (GTO): proprioceptor located at the junction between a muscle and its tendon, running **in series** with the extrafusal muscle fibres. It is innervated by sensory (Ib afferent) neurons. **Function:** Detects **muscle tension** (force) and the rate of change of tension. It is the primary receptor for the **Golgi tendon reflex** (inverse stretch reflex). This is a **protective reflex** to prevent muscle tearing.

6. Stretch Reflex (Myotatic Reflex): A monosynaptic (one synapse) spinal reflex where stretching a muscle causes it to contract. **Example:** The **knee-jerk reflex** (patellar tendon tap). **Clinical Importance:** Hyperactive or hypoactive stretch reflexes are key clinical signs for localizing neurological damage.

7. **Golgi Tendon Reflex (Inverse Stretch Reflex):** A polysynaptic spinal reflex where excessive muscle tension causes the muscle to relax. **Example:** A weightlifter dropping a load that is too heavy – the muscle relaxes to prevent tendon tearing. It also activates **reciprocal excitation** (the antagonist muscle contracts).

8. **Withdrawal Reflex (Flexor Reflex):** A polysynaptic, protective spinal reflex that causes rapid withdrawal of a body part from a painful or noxious stimulus.

9. **Crossed Extensor Reflex:** A spinal reflex that often occurs simultaneously with the withdrawal reflex to maintain **posture and balance**. It coordinates both limbs to prevent falling.

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